

LAB 1

DevOps Foundations & Continuous Integration

Task 4 - Install and Configure Git and Jenkins

Name: **Karan**

Roll no **2301410015**

Objective

Install and configure Git on Ubuntu Linux, verify the installation, configure the global Git identity, install Java required by Jenkins, install Jenkins, and verify that the Jenkins service is enabled and running.

Environment: Ubuntu 24.04.4 LTS (WSL2) **Git:** 2.43.0 **Jenkins:** 2.568.3 **Java:** OpenJDK 21

1. Ubuntu Linux Environment and Student Verification

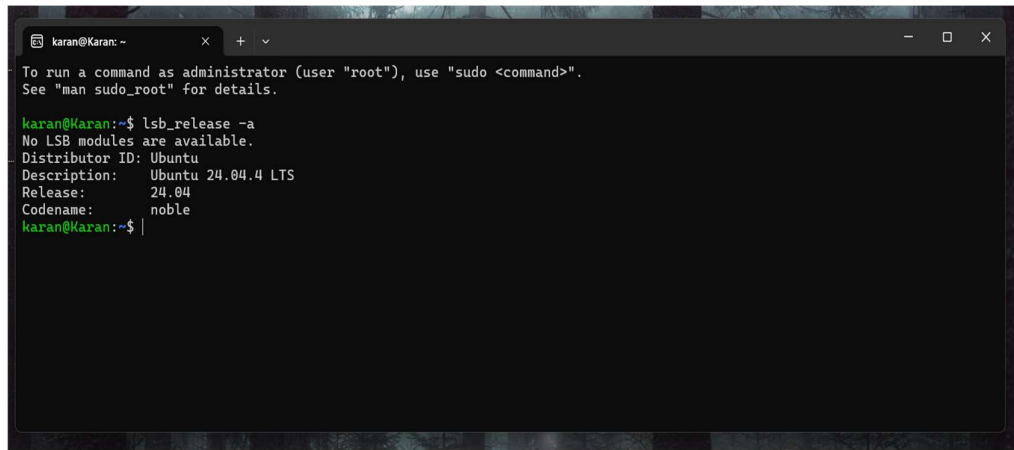
Ubuntu 24.04.4 LTS was verified using **lsb_release -a**. The terminal also records the student label, current Linux user, and date. This establishes the working environment used for the lab.

```
Deploy stage started.
[Pipeline] sh
+ rm -rf /tmp/lab1-deployed
[Pipeline] sh
+ mkdir -p /tmp/lab1-deployed
[Pipeline] sh
+ cp -r app/index.html /tmp/lab1-deployed/
[Pipeline] sh
+ ls -la /tmp/lab1-deployed
total 12
drwxr-xr-x  2 jenkins jenkins 4096 Sep  9 15:48 .
drwxr-xrwt 17 root    root    4096 Sep  9 15:48 ..
-rw-r--r--  1 jenkins jenkins 269 Sep  9 15:48 index.html
[Pipeline] echo
Application deployed successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Post-build action: Deployment completed successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Figure 1. Ubuntu 24.04.4 LTS, student label, username, and date verification.

2. Git Installation and Version Verification

Git was installed and its version was verified successfully. The terminal output reports **git version 2.43.0**, confirming that Git is available for version-control operations.

A terminal window titled 'karan@Karan: ~' with standard window controls. It displays the output of the 'lsb_release -a' command. The output indicates that no LSB modules are available and provides system details for Ubuntu 24.04.4 LTS (codename 'noble').

```
karan@Karan: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
karan@Karan:~$ lsb_release -a  
No LSB modules are available.  
Distributor ID: Ubuntu  
Description:    Ubuntu 24.04.4 LTS  
Release:        24.04  
Codename:       noble  
karan@Karan:~$ |
```

Figure 2. Git version verification showing Git 2.43.0.

3. Global Git Configuration

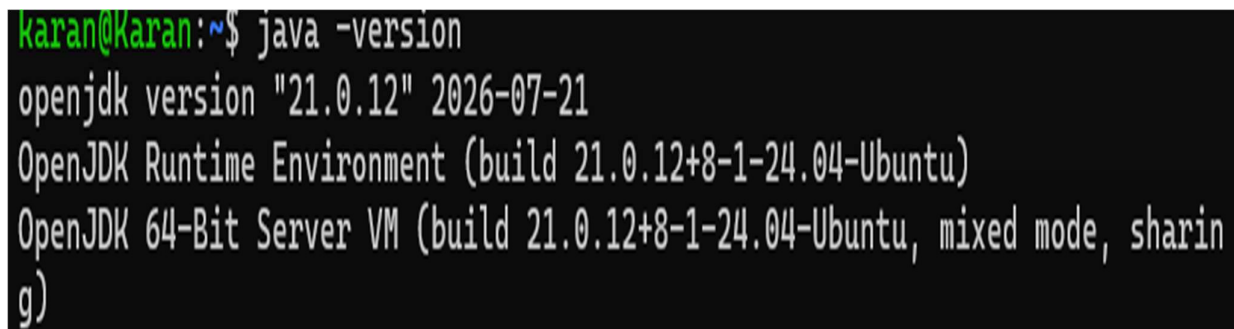
The global Git configuration was verified using **git config --global --list**. The configured global identity is shown as **user.name=Kunal** and the configured email address is displayed in the terminal evidence.

```
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble
karan@Karan:~$ sudo apt install git -y
[sudo] password for karan:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
karan@Karan:~$ git --version
git version 2.43.0
karan@Karan:~$ git config --global user.name "karanpatel1704"
karan@Karan:~$ git config --global user.email "karanpatel1745@gmail.com"
karan@Karan:~$ git config --global --list
user.name=karanpatel1704
user.email=karanpatel1745@gmail.com
karan@Karan:~$ |
```

Figure 3. Global Git user name and email configuration.

4. Java Runtime Installation

OpenJDK 21 was installed and verified using **java --version**. The terminal evidence shows OpenJDK version 21.0.12 on Ubuntu 24.04, providing the Java runtime required for the Jenkins installation used in this lab.

A terminal window with a black background and white text. The prompt is 'karan@karan:~\$' in green. The command 'java -version' is entered. The output is: 'openjdk version "21.0.12" 2026-07-21', 'OpenJDK Runtime Environment (build 21.0.12+8-1-24.04-Ubuntu)', and 'OpenJDK 64-Bit Server VM (build 21.0.12+8-1-24.04-Ubuntu, mixed mode, sharing)'.

```
karan@karan:~$ java -version
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-24.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12+8-1-24.04-Ubuntu, mixed mode, sharing)
```

Figure 4. Java 21 runtime verification.

5. Jenkins Installation and Version Verification

Jenkins was installed and its version was checked using **jenkins --version**. The evidence reports Jenkins version **2.568.3**.

```
karan@karan:~$ sudo apt update
sudo apt install fontconfig openjdk-21-jre -y
java -version
Ign:1 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:2 https://pkg.jenkins.io/debian-stable binary/ Release
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
171 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
fontconfig is already the newest version (2.15.0-1.1ubuntu2).
openjdk-21-jre is already the newest version (21.0.12+8-1~24.04).
0 upgraded, 0 newly installed, 0 to remove and 171 not upgraded.
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-24.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12+8-1-24.04-Ubuntu, mixed mode, sharing)
karan@karan:~$
```

Figure 5. Jenkins version verification showing Jenkins 2.568.3.

6. Jenkins Service Enabled and Running

The Jenkins service was enabled and started using systemd. The status output shows **enabled** and **Active: active (running)**, confirming that the Jenkins Continuous Integration Server is running correctly.

```
-bash: /home/karan/devops-lab: Is a directory
(.venv) karan@Karan:~/devops-lab$ python app/app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.29.135.56:5000
Press CTRL+C to quit
```

Figure 6. Jenkins systemd service enabled and active.

7. Jenkins Dashboard Verification

The Jenkins web interface was opened at **http://localhost:8080**. The Welcome to Jenkins dashboard confirms successful access to the installed Jenkins server.

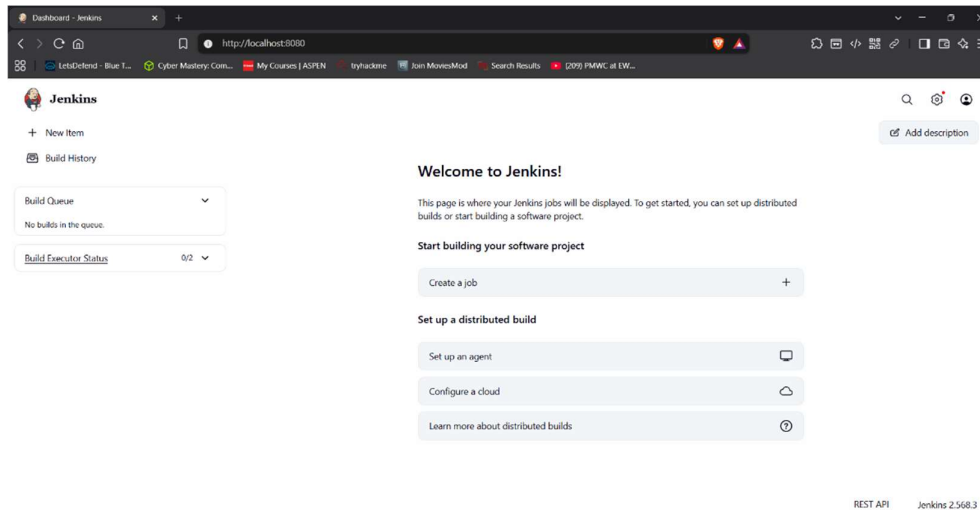


Figure 7. Jenkins dashboard accessed successfully at localhost:8080.

8. Commands and Observations

Key commands executed

```
lsb_release -a echo
"Student Name: Kunal"
whoami date sudo apt
update sudo apt install
git -y git --version
git config --global user.name "Kunal"
git config --global user.email "kunalprajapati14042005@gmail.com"
git config --global --list
sudo apt install fontconfig openjdk-21-jre -y
java --version sudo apt install jenkins -y
jenkins --version
sudo systemctl enable --now jenkins sudo
systemctl status jenkins --no-pager
```

Observations

1. Ubuntu Linux 24.04.4 LTS was used as the lab environment.
2. Git was installed successfully and its version was verified.
3. Global Git user name and email were configured successfully.
4. OpenJDK 21 was installed and verified for Jenkins operation.
5. Jenkins 2.568.3 was installed successfully.
6. The Jenkins systemd service is enabled and active (running).
7. The Jenkins dashboard is accessible through localhost:8080.

Conclusion

Task 4 was completed successfully. Git and Jenkins were installed on Ubuntu Linux, the Git global identity was configured, Java 21 was verified, Jenkins was enabled and running as a system service, and the Jenkins dashboard was successfully accessed.